Huawei Access Node KOR-ACC-83 Optical Transport Failure Resolution Procedure (ALARM_2933)

Node: KOR-ACC-83 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_2933

Description

The occurrence of an Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Huawei Access Node KOR-ACC-83 at the Koramangala Datacenter can be attributed to the inherent vulnerabilities of the Optical Transport Network (OTN) architecture. In this specific role, the Huawei Optical Networking Platform (ONGP) utilizes the Synchronous Digital Hierarchy (SDH) and OTN protocols to transmit data over fiber-optic cables. The SFP/XFP degradation, fiber cuts, or Rx/Tx power loss can be caused by factors such as aging of the optical fibers, mechanical stress on the fiber-optic cables, or hardware malfunctions within the SFP/XFP modules. The Huawei Access Node KOR-ACC-83, being part of the ONGP, relies on the integrity of these optical fibers and hardware components to maintain its optical transport functionality. Therefore, any failure in this domain necessitates immediate attention to prevent network disruptions and data losses.

Pre-requisite Checks

- Verify that the remote hands are dispatched to the Koramangala Datacenter.

- Check the status of the redundant link to ensure that it is operational and can take over in case of a failure.

- Confirm that all necessary permission and authorization are granted to perform the necessary actions.

Potential Root Causes

Cause	Probability
Aging of the optical fibers	High
Mechanical stress on the fiber-optic cables	Medium
Hardware malfunctions within the SFP/XFP modules	Low

Diagnostic Commands & Expected Outputs

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Command: display otn sfp, Expected Output: SFP information: Module 1, Type: SFP-10G-SW-LC, Status: OK, Description: This command checks the status of the SFP module and verifies if it is functioning correctly.

Command: display otn fiber, Expected Output: Fiber information: Fiber 1, Type: SMF, Status: OK, Length: 10km, Description: This command checks the status of the fiber-optic cable and verifies if it is functioning correctly.

Command: display otn rx-tx, Expected Output: Rx/Tx information: Module 1, Rx Power: -20dBm, Tx Power: -5dBm, Description: This command checks the Rx/Tx power levels of the SFP module and verifies if they are
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within the specified range.
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Resolution Steps

Step 1: Step: Replace the faulty SFP module, Command: replace otn sfp module 1, Description: This step replaces the faulty SFP module with a new one to restore the optical transport functionality.

Step 2: Step: Repair or replace the fiber-optic cable, Command: repair otn fiber 1, Description: This step repairs or replaces the faulty fiber-optic cable to restore the optical transport functionality.

Step 3: Step: Adjust the Rx/Tx power levels, Command: adjust otn rx-tx power module 1, Description: This step adjusts the Rx/Tx power levels of the SFP module to ensure that they are within the specified range.

Post-Check Verification

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Command: display otn sfp, Expected Output: SFP information: Module 1, Type: SFP-10G-SW-LC, Status: OK, Description: This command verifies that the SFP module is functioning correctly after the resolution steps.

Command: display otn fiber, Expected Output: Fiber information: Fiber 1, Type: SMF, Status: OK, Length: 10km, Description: This command verifies that the fiber-optic cable is functioning correctly after the resolution steps.

Command: display otn rx-tx, Expected Output: Rx/Tx information: Module 1, Rx Power: -20dBm, Tx Power: -5dBm, Description: This command verifies that the Rx/Tx power levels of the SFP module are within the specified range after the resolution steps.
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Roll-back Procedures

- Step: Restore the original SFP module, Command: restore otn sfp module 1, Description: This step restores the original SFP module to its previous state in case the replacement does not resolve the issue.

- Step: Restore the original fiber-optic cable, Command: restore otn fiber 1, Description: This step restores the original fiber-optic cable to its previous state in case the repair or replacement does not resolve the issue.

- Step: Restore the original Rx/Tx power levels, Command: restore otn rx-tx power module 1, Description: This step restores the original Rx/Tx power levels of the SFP module to its previous state in case the adjustment does not resolve the issue.